DATA & CATALOG AUDIT · SIMPLE LANGUAGE · **REV 1.1**

Dish Registry Audit Report

What lives in `dish_registry.json` and `el_dorado_folsom_culinary_matrix.json`, which fields the app actually uses, how Postgres tables fit in, and whether prod/staging databases are awake. **Rev 1.1** adds Ask-match / F() holes (soak: "Butter chicken" → Butter Dosai), refreshes identity coverage after ROE-022 enrich, and re-prioritizes the fix backlog so math tickets cannot paper over bad match rules.

REGISTRY	MATRIX	LIVE INDEX	PROD DB
~3.9 MB · 2,772 city rows	~0.7 MB · 13 EDH/Folsom kitchens	generatedAt 2026-09-06	Active (not paused)

STAGING DB	AUDIT DATE
Active · other org	6 Sep 2026 · Rev 1.1

Read order: this audit (findings) → `docs/rasaoi-dish-matrix-recommendations.pdf` Rev 1.1 (phased fixes). Do not jump to J(r) / `score-reading` until P0 match-rule items below are closed or ticketed.

CONTENTS

01 Snapshot (what matters)

2 sources → 1 slim index
Raw JSON never ships in the SPA

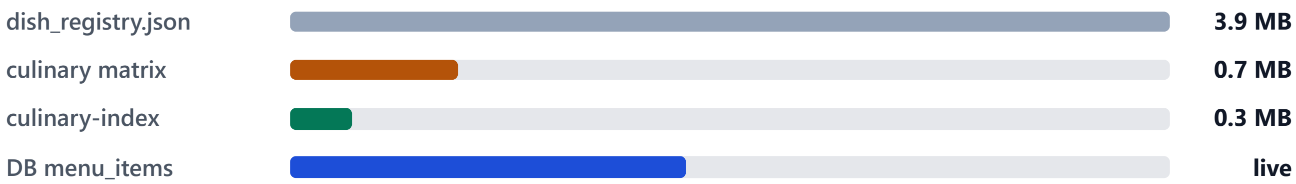
DB = live plates
menu_items is the catalog gate

Matrix = enrich
Price, macros, identity hints

India registry mostly idle
Only EDH/Folsom slice builds into index

Core idea: restaurants in Postgres decide *what can be plated*. The culinary index helps rank and explain. The giant India registry is a research book — not the live menu.

Where dish facts live (relative weight)



File sizes measured on disk 6 Sep 2026. DB size varies by seeded venues.

02 Database status (prod + staging)

Checked without using secret keys. A **paused** Supabase project usually returns a clear "project paused" message. Both projects instead answer on the API gateway with **HTTP 401** (need a key) — that pattern means the project is **up / not paused**.

ENV	PROJECT REF	NAME (CLI)	PROBE	STATUS	NOTES
Production	`kiugplotjcnmpwjlxajc`	rasaoi-project	REST/Auth → 401	ACTIVE	Listed in logged-in CLI org. Vercel prod target.
Staging	`aotlzhdgnvovvqxmgyyx`	rasaoi-staging	REST/Auth → 401	ACTIVE	Active API. CLI org for personal login does **not** own this project (secrets/deploy 403). Dashboard needs org `begvkmrmyqxrrrtoqurj`. Preview: v0-rasaoi-staging / rasaoi-i8.

HOW WE DECIDED "ACTIVE"

No "Project paused" body. Gateway responds. Auth endpoint requires a key (401), not a pause page.

WHAT WE DID NOT CLAIM

Row counts / billing plan. Those need anon or service keys (not printed in this audit).

Local `.env` currently targets **staging** (`aotlzhdgnvovvqxmgyyx`), not prod. Linked CLI project matches staging. Do not assume laptop queries hit prod.

03 Three data worlds

WORLD	FILES / TABLES	JOB	TOUCHES READING?
A. Research registry	`dish_registry.json`	City×dish price / link / nutrition book (mostly India + local USD slice)	Only EDH/Folsom slice → index
B. Local culinary matrix	`el_dorado_folsom_culinary_matrix.json`	Per-kitchen course tree (appetizer → carrier) for EDH/Folsom	Yes, via build → index
C. Live catalog (DB)	`restaurants.menu_items` + `dishes`	What Lab committed; plate gate + ranking inputs	Yes — primary truth for plates

```
dish_registry.json ─┐
                    │  build-culinary-index.mjs (offline, no AI)
el_dorado_folsom_…json ─┤
                    ▼
       src/data/culinary-index.json  ──▸  veda / pairings / glycemic / askFulfillment
                    ▲
                    │  optional overlay (staging flag)
       experimental_dish_knowledge (staging Postgres)

Separate path:
  Lab → ingest-menu → commit-dishes → restaurants.menu_items + dishes
  Reading plate gate: menu_items ∪ matrix/index — never invent
```

04 Registry field map (`dish_registry.json`)

Top shape: `{ by_city, nutrition }`. Keys in `by_city` look like `dish-slug|city-slug`.

4.1 `by_city` entry fields

FIELD	MEANING	USED IN BUILD?	USED AT SCORE TIME?	HEALTH
`name`	Dish display name	Yes (local only)	Via index dish key	USED
`city`	City label	Local filter	No	FILTER ONLY
`restaurant`	Venue name (local rows)	Yes → restaurant bucket	Lookup key	USED (LOCAL)
`price.amount / currency`	Listed price	USD → priceUsd / median	Budget soft signals	INR IGNORED
`price.source_url / verified_at / fetch_status`	Provenance	Dropped in slim	No	AUDIT ONLY
`link.url / platform`	Zomato / Grubhub link	Dropped	No	UNUSED IN APP
`nutrition.*`	Macros / diabetic note	Slim macros if present	GL / heaviness	SPARSE GLOBALLY
`taste_benchmark.*`	Rating idea	Dropped	No	100% EMPTY SCORE
`enriched_at`	Enrich timestamp	Dropped	No	OPS ONLY
`source_matrix`	Marks EDH matrix rows	Local include flag	No	USEFUL FILTER

4.2 `nutrition` map (global by dish slug)

FIELD	SLIMMED INTO INDEX?	NOTES
`calories_kcal`, `protein_g`, `fiber_g`	Yes → `byDish`	Fallback when restaurant dish thin
`diabetic.gi_band`	Only if string set	Almost never set today
`overall`, `source_url`, `fetch_status`	No	Research metadata

2,772
by_city rows

727
EDH/Folsom (USD) local

2,045
INR (India) rows

2,304
global nutrition keys

0
taste scores filled

1,876
by_city with nutrition=null

05 Matrix field map (`el_dorado_folsom_culinary_matrix.json`)

Tree: `proteinFamily → region → metro → city → [ { menu_style, courses } ]`.

FIELD PATH	MEANING	BUILD KEEPS?	RUNTIME CONSUMER	HEALTH
Tree key (`veggies|meat|…`)	Protein family root	`proteinFamilies[]`	askFulfillment (fallback)	CAN MISLEAD VS IDENTITY
`menu_style`	Kitchen name	Restaurant key + alias	lookupRestaurant	CRITICAL
`courses.*.name`	Dish in that slot	Dish record	Plates / fulfill / GL	CRITICAL
`courses.*.price`	USD amount + provenance	`priceUsd` only	Budget / display enrich	USED (AMOUNT)
`courses.*.link`	Grubhub etc.	Dropped	No	UNUSED
`courses.*.nutrition`	Macros + method	Slim macros + dish_type	veda heaviness, glycemic	CALORIES PRESENT
`nutrition.diabetic.gi_band`	GI band	If set	glycemic.ts	0% FILLED
`courses.*.taste_benchmark`	Rating	Dropped	No	EMPTY
`courses.*.identity`	ROE-019 proteins / diet / role	Prefer over infer	askFulfillment	0 IN SOURCE FILE

8
protein families

13
unique kitchens

416
course dish slots

104×4
app / starter / main / base

416
with calories

0
with gi_band / identity in file

> Appetizer and starter often repeat the same dish name. Build merges by normalized name and keeps the richer nutrition row — but course labels can blur.

06 Flow: how fields get touched

Ask / dials
parse-intent
→
Postgres menus
menu_items names
→
culinary-index
price · macros · identity
→
veda score
rank kitchens
→
pairings
triple plates

Field touch points

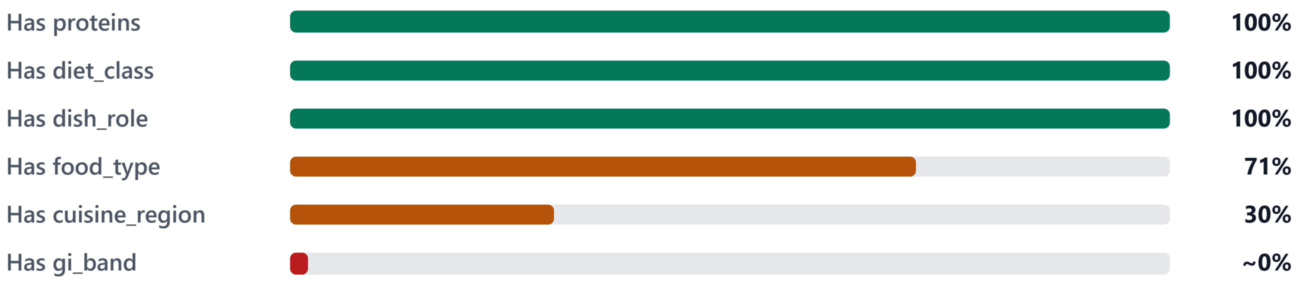
name → dish match / catalog gate · priceUsd → budget tilt · calories/dish_type → heavy/light · proteins/diet_class/food_type → Ask-fulfillment · gi_band → blood-sugar lens

Not touched at score time: link, taste_benchmark, INR prices, India-only by_city rows, raw source_url provenance

RUNTIME MODULE	INDEX / DB FIELDS IT READS	WHY
`veda.ts`	menu names; lookupRestaurant/Dish; dish_type; calories	Rank kitchens; dish hit; heaviness tags
`askFulfillment.ts`	identity proteins / diet / food_type / dish_role; proteinFamilies fallback; menu ∪ matrix names	Ask-first ranking (ROE-019)
`pairings.ts`	menu_items; matrix dishes; carriers; catalog gate	Best / Clean / Heritage + carrier
`glycemic.ts`	gi_band; calories; dish_type; else edge estimate	Blood-sugar lens
`catalogGuard.ts`	menu ∪ matrix membership	Drop invented plates
Lab / `commit-dishes`	Writes `dishes` + rebuilds `menu_items`	Live catalog updates (no matrix write)
Staging overlay	`experimental_dish_knowledge`	Optional dynamic enrich when flag on

07 Coverage & quality bars

Built index dish fields (156 scored dishes · post ROE-022 enrich)



Rev 1.1: identity enrich (ROE-022 T4) raised proteins/diet/role from ~20% → ~100%. **Still does not fix named-dish token match** (§9b). Index `generatedAt`: 2026-09-06. Remaining gaps: cuisine_region ~30%, gi_band ~0%, food_type ~71%.

Registry usefulness for EDH product

~74% of by_city rows never enter the culinary index (non-local filter).

08 Postgres tables (prod schema + staging extras)

8.1 Production tables

TABLE	KEY COLUMNS	PUBLIC RLS	LINK TO REGISTRY/MATRIX
`restaurants`	name, cuisine, purity/oil/grain, `menu_items` JSONB, signature_dish, phone/address, promos URLs	SELECT	Names should align with matrix `menu_style` via aliases — not auto-synced
`dishes`	name, diet_class, dietary_*, oil/grain, glycemic_load, price, cooking_method, cuisine_region…	SELECT	Lab graph; richer than matrix; not the same as culinary-index
`restaurant_sources`	source_url, scraped_at, parse_confidence	SELECT	Ingest provenance (like matrix link, but for Lab)
`active_promos`	restaurant_id, deal fields	SELECT	None
`outcome_selections`	dish, restaurant, dials, check-in fields	Insert only	None (telemetry)
`dishes_feedback`	thumbs, dials_snapshot	No public	Lab QA

8.2 Staging-only experimental tables

TABLE	PURPOSE	MAPS CLOSEST TO
`experimental_dish_knowledge`	Dynamic dish facts + optional embedding	Slim culinary-index row
`experimental_nutrition_quarantine`	Unmapped ingredients	Failed nutrition enrich
`experimental_llm_semantic_cache`	Intent cache	—
`experimental_outcome_feedback`	Learning sandbox	—

> Prod does **not** store the culinary matrix or dish_registry. Those stay in git → compiled JSON (and staging knowledge table when flagged).

```
Prod Postgres                  Git / build                 Staging only
restaurants.menu_items  ←— Lab      matrix + registry —→ culinary-index.json
dishes (attributes)                 (offline build)         experimental_dish_knowledge
active_promos / outcomes                                    quarantine / cache
```

09 Issues + remediations

ISSUE Giant India registry (~74% of rows) is unused at product score time but costs repo weight and agent confusion.

IMPACT Slow clones; people may think INR/Zomato data drives EDH Readings.

FIX Split `dish_registry.local.json` (EDH/Folsom) from archive `dish_registry.india.json`. Build script reads local only. Document clearly in CONTEXT_PLAN.

ISSUE `taste_benchmark.score` is always null; links/provenance dropped at build.

IMPACT Dead schema noise; false hope of social proof from registry.

FIX Either populate from a trusted source with tier labels, or remove from product schema and keep only in research notes.

ISSUE Matrix source historically had sparse `identity`; `gi_band` still ~null; `cuisine_region` only ~30% on index after ROE-022.

IMPACT Protein Asks improved via enrich; GL lens still Edge-heavy; regional integrity still incomplete.

FIX Keep enrich in CI; finish cuisine_region per kitchen; backfill gi_band where method known; Vitest soak. **Separate:** match-rule P0 in §9b — identity alone does not fix "Butter chicken" → Butter Dosai.

ISSUE Dual catalogs: DB `menu_items` vs matrix kitchen list can drift (aliases, retired venues, new Lab commits).

IMPACT Enrich miss → weaker F(); or matrix dish without DB membership (catalog gate drops / honest miss).

FIX Add CI check: every matrix `menu_style` resolves to a DB restaurant (or allowlist). Lab commit optionally refreshes matrix stub. Alias table in one place.

ISSUE Appetizer/starter duplication doubles slots (416) for ~half unique mains conceptually.

IMPACT Noisy merges; course semantics unclear.

FIX Collapse to `starter + main + carrier` in next matrix revision; migrate builder COURSE_KEYS.

ISSUE Protein tree root (`meat`/`veggies`) can disagree with real dish proteins when identity missing.

IMPACT Classic soak: chicken-only kitchen ranks high for "meat not murgi".

FIX Trust `identity.proteins` over tree root (already coded). Coverage now ~100% proteins/diet after ROE-022 — maintain via enrich CI. Remaining soak misses → Lab tags, not screenshot hotfixes.

ISSUE Index can lag matrix/registry edits if rebuild forgotten (mitigated when generatedAt is current).

IMPACT Stale enrich on Reading until redeploy.

FIX CI job: if matrix/registry hash changes, fail unless culinary-index rebuilt. Longer term: live culinary cache (combined upgrade proposal).

ISSUE Staging knowledge path vs prod static index — two mental models; staging Gemini/org access is separate from catalog.

IMPACT "Works on staging" enrich missing on prod; staging secrets may be unreachable from personal CLI.

FIX Promotion checklist: static rebuild for prod until live cache ships; never auto-promote Apify into `menu_items`; document staging org ownership.

`09b` Ask-match / F() holes (Rev 1.1)

Catalog + identity answer "what exists." **Match rules** answer "does this plate fulfill the Ask?" Rev 1.0 and the matrix recommendations treated named-dish Asks as *low confusion*. Personal soak **6 Sep 2026** shows that assumption is wrong while token match stays soft.

> **Soak evidence:** Ask `Butter chicken under 40` → hero **Mylapore 100%** with **Butter Dosai** (veg tiffin). **India Oven** (exact Butter Chicken) sat in Alternatives. Parse pill showed BUTTER CHICKEN — Gemini was fine; offline F() / ranking failed.

HOLE ID	WHERE	FAILURE MODE	WHY IDENTITY/MATH DON'T CLEAR IT
`M-01`	`namedDishMatchStrength`	Multi-token Ask ("butter"+"chicken") scores **partial** on shared garnish/fat token alone	Identity proteins on Dosai don't remove the butter token hit
`M-02`	ROE-018 honesty in `veda.ts`	~100% / "No exact dish" only when `dishMatch === "none"`; weak partial still looks like a win	J weights scale the same bad F signal
`M-03`	`venueAskFulfillment`	`alignedCount ≥ 2` weak partials can promote level to **full** ("Ask fulfilled")	Moving F to Edge (ROE-024) copies the bug unless rules change
`M-04`	Sort: fulfillment tier then score	When F tiers tie, purity / Sovereign / oil tags can crown the wrong kitchen	Pareto (ROE-025) diversifies slots among already-wrong winners
`M-05`	Protein Ask path vs named path	Preferred protein from tokens may not *veto* plates that only matched a non-protein token	Enrich helps "meat not chicken"; not butter-collision named dishes
`M-06`	Stop-word / synonym policy	Cooking fats (`butter` / `ghee`) are real dish tokens on South tiffin names — no "require protein when present" rule	Tags/DB freshness don't redefine token policy

Expected correct behavior (acceptance sketch)

- Named Ask with a protein token → exact/strong F only if that protein (or synonym) appears on the plate/catalog line.
- Butter-only overlap → at most "Closest" / capped confidence; never naked ~100% hero.
- Kitchens with exact named dish outrank kitchens with garnish-only overlap (Ask-fulfillment first).
- Vitest soak: `Butter chicken`, `Butter chicken under 40`, `ghee roast` (legitimate butter/ghee dish), `Mysore pak` (no false Mysore dosa).

> **Ticket posture:** treat M-01…M-06 as **P0 / next free ROE** (or explicit scope on ROE-023), *before* relying on score-reading / Pareto to "fix Readings." Documented remediations feed `rasaoi-dish-matrix-recommendations` Rev 1.1 §03 / §06 / §08.

10 Fix backlog (priority) — Rev 1.1

#	REMEDIATION	EFFORT	PRIORITY	PERF / QUALITY WIN
0	**P0 match rules:** tighten `namedDishMatchStrength` + F() (M-01…M-06); honesty for weak partial; Vitest soak Butter chicken / Mysore pak / ghee roast	MED	P0	Named-dish hero correctness — blocks false 100%
1	CI: rebuild gate when matrix/registry change	LOW	P1	Stops silent stale index
2	Finish cuisine_region + food_type gaps; maintain proteins/diet enrich in CI (post ROE-022)	LOW–MED	P1	Regional + type Ask signal
3	Alias / name parity check matrix ↔ DB (ROE-022 T5 shipped — keep green)	LOW	P1	Fewer enrich misses
4	Split local vs India registry files	MED	P2	Repo clarity; smaller ops surface
5	Backfill gi_band where cooking method known; quarantine rest	MED	P2	Fewer Edge GL calls; clearer lens
6	Drop or quarantine empty taste_benchmark from product path	LOW	P3	Less schema noise
7	Collapse appetizer/starter in matrix vNext	MED	P3	Cleaner course semantics
8	Live culinary cache + `score-reading` (combined upgrade) — *after* P0 match rules	HIGH	P1 (architecture)	Freshness + payload — must carry fixed F()

> **Keep:** offline build (no AI at score time), catalog plate gate, Lab review before commit, staging experimental tables off prod, dish-non-invention, Ask-fulfillment first *with honest F()*.